Time Series X- and Ku-Band Ground-Based Synthetic Aperture Radar Observation of Snow-Covered Soil and Its Electromagnetic Modeling

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***Abstract*— The snow water equivalent (SWE, a measurement of the amount of water contained in snow packs) is an important variable in earth systems. Microwave remote sensing provides a possible solution for estimating the SWE globally. To support radar SWE retrieval, the snow backscattering theory needs to be studied; the forward simulation model needs to be validated against natural snow observations. In this study, a one-winter experiment to observe the time series backscattering coefficient of snow-covered bare soil is reported. This is the first long time series snow-covered soil backscattering experiment that was measured by an imaging radar. The backscattering coefficient was observed at three frequencies covering the X-band and dual-Ku bands, which are of great interest to the snow remote sensing community and are used for SWE estimation in mountains. The calibration of the synthetic aperture radar (SAR) system was conducted manually and carefully to ensure high-quality radar observation data. The observations from our experiment show that in general, the time series backscattering signature of snow-covered terrain is mainly driven by soil freezing, snow grain size growth, and snow accumulation processes. The time series observations for dry snow are modeled by backscattering models with model inputs directly calculated from field measurements. Our simulation results indicate that the time series radar backscattering at three frequencies and four polarizations can be simulated with high accuracy, including the cross-polarization channels. This study provides some key understanding of the time series signature of radar backscattering from snow and provides some key implications for SWE retrieval from radar observations.**

***Index Terms*— Electromagnetics for remote sensing, ground experiment, radar remote sensing, snow.**

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I. Introduction

THE snow water equivalent (SWE, a measurement of the amount of water contained in snow packs) is an important variable in earth systems. Detailed knowledge of the SWE is crucial for meteorology, hydrology, water resource, water cycle. and global change studies. Eighty percent of the land in Eurasia and North America is covered with snow during the winter. Approximately 46.5 million km^2 of snow covers the Northern Hemisphere in January and 3.8 million km^2 in August [15]. *In situ* sampling of the SWE is difficult and sparse in space and time because of the harsh environment and meteorological conditions of snow-covered areas. The SWE can be simulated using land surface process models or snow process models, but large uncertainties spatially and temporally exist and are affected by both the forcing data and the model validity [16].

Microwave remote sensing provides a possible solution to estimate the SWE. The penetration of the microwave signal to the snow layer makes the SWE and snow depth retrieval from microwave remote sensing possible. Since the 1980s, passive microwaves have been applied for large-scale and coarse-resolution mapping of the SWE. Due to the coarse resolution of passive microwave data and the saturation of passive microwave signals at high frequencies for deep snow, its application in mountainous regions was not successful unless supported by data assimilation techniques (Li *et al.* [17]).

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However, seasonal snow in mountainous regions is of high quantity and long duration and is a critical part of the global SWE. Therefore, to map the mountainous SWE, an effective remote sensing method with high spatial resolution is needed. Active remote sensing can achieve much higher spatial resolution than passive microwave remote sensing. Thus, many studies have been conducted to study the feasibility of SWE remote sensing by radar. One of the basic concepts of SWE remote sensing by radar is based on the interaction of radar signals with snow volume. An SWE retrieval algorithm was first proposed by Shi and Dozier [13], [14] using spaceborne imaging radar-C (SIR-C)/X-band synthetic aperture radar (SAR) L-, C-, and X-band SAR backscattering data. It was then suggested that higher frequency X- and Ku-band radar is more suitable for SWE retrieval. In the Cold Regions Hydrology High-resolution Observatory (CoReH2O) project, a candidate mission of the ESA Earth Explorer Programme [18], [19], a dual-frequency SAR operating at the Ku-band (17.2 GHz) and X-band (9.6 GHz) frequencies was proposed.

To support radar SWE retrieval, the snow backscattering theory needs to be studied; the forward simulation model needs to be validated against natural snow observations. In a phase-A study of the CoReH2O mission, a tower-based scatterometer and airborne SAR (see [18], Rott *et al.* [20]) were used to measure X- and Ku-band polarimetric backscattering from snow-covered soil. The experimental data serve as a data source for the validation of backscattering models and retrieval algorithms. The sensitivity of radar backscattering to SWE is the basis for SWE retrieval from radar backscattering intensity. However, previous studies have proven that the microstructure (or simply speaking, grain size) of snow still plays an important role in the radar backscattering of snow-covered soil in the X- and Ku-bands [19], [21]. Thus, to better understand the interaction of microwave signals with snow volume and the snow–soil system, a dedicated ground-based experiment needs to be designed. Although ground-based radar measurements of snow-covered terrain have been conducted in several studies [2]–[12], few experiments have been conducted in an intensive time series manner [4], [48], and such intensive radar measurements are extremely rare for snow in China. The advantage of a time series measurement is that it can cover both small and large snow grain sizes during snow grain growth processes. This temporal variation can be much stronger than a spatial variation of hundreds of kilometers. The radar can be installed at a fixed location and is easier to operate and maintain. This guarantees a successful experiment with a lower transportation cost.

In this study, we designed and conducted a one-winter ground-based radar experiment for snow-covered terrain from October 2017 to March 2018 at the Altay National Meteorological Station (88° 05′ E, 47° 44′ N) in Xinjiang, China, from October 2017 to March 2018. The X- (10 GHz) and Ku- (14 and 17.75 GHz) band polarimetric backscattering coefficients at VV, VH, HH, and HV polarizations were measured using a ground-based SAR (GB-SAR) system. In this system, a horn antenna, the size of which differs at different frequencies, was installed on a 3.5-m rail and moved along the rail to provide a high-resolution backscattering map of a 10 m × 34-m observing field. The system can measure one frequency at a time. Compared with previous experiments, we calibrated the radar system much more frequently, usually immediately before or after the snow measurement within 2 h and each time when the antenna was changed. Careful absolute calibration was performed to ensure high precision, checked by the achieved standard signal of a medium-size trihedral. In this case, the calibration of backscattering should be very reliable on the temporal scale. The polarimetric backscattering of snow-covered soil is determined by many snow and soil parameters, including the snow depth, snow microstructure, snow density, snow temperature and their vertical stratification, soil temperature, soil moisture, and soil roughness. All these parameters were measured in the field within ∼50 m of the radar measurements. Two fully physically based backscattering models were validated and evaluated. Experimental and theoretical approaches are combined to understand microwave scattering interactions with snow-covered terrain. We address the following questions in this study.

1) What are the time series signatures of X- and Ku-band snow-covered terrain backscattering? What are the driving factors of the time series backscattering evolution of snow-covered terrain in the X- and Ku-bands?
2) Can the backscattering models simulate the X- and Ku-band polarimetric backscattering of snow-covered terrain with high accuracy simultaneously at all channels, especially the cross-polarization channels?
3) Can the field-measured snow parameters be used directly in backscattering models?

II. Experimental Data Collection

From October 2017 to March 2018, a GB-SAR experiment for snow and frozen soil was carried out at the Altay National Meteorological Station (88° 05′ E, 47° 44′ N). The X-band (center frequency 10 GHz) and dual Ku-band (center frequency 14 and 17.75 GHz) polarimetric backscattering coefficients were measured using a GB-SAR system. Two microwave radiometers were also deployed to measure the brightness temperature. The experiment setup is shown in Fig. 1. The GB-SAR observation field was a flat open field with all vegetation carefully removed before the first snowfall. The snow depth, snow density, and snow grain size were measured frequently near the GB-SAR measurement field without disturbing the SAR imaging area.

A. GB-SAR Measurement

The GB-SAR system used for backscattering measurement is based on a vector network analyzer (VNA), which has a frequency range of 0.5–20 GHz. Four standard horn antennas are used to measure VV, HH, VH, and HV backscattering signals, which are mounted on a 3.5-m rail. The antennas move along the rail to form a 3.5-m synthetic aperture, which is driven by a motor with a positioning precision of 0.01 mm. A microwave switch is used to switch between antennas of different polarizations used to transmit and receive microwave signals to form VV, HH, VH, and HV backscattering measurements. GB-SAR is a stepped frequency continuous wave

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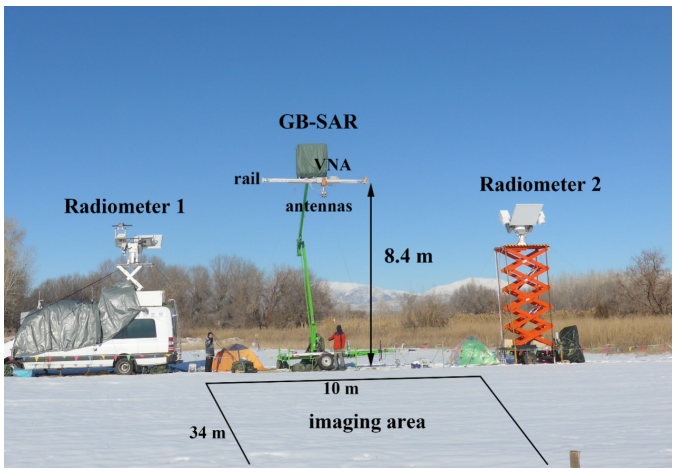


Fig. 1. GB-SAR setup in field.

(SFCW) radar, and the received complex signal values in the frequency domain are recorded as the raw data. In SAR scanning, a signal with 30-dBm power is transmitted so that a very low noise level can be achieved.

In this experiment, the polarimetric backscattering coefficients of three center frequencies at the X- and Ku-bands were measured, which are 10, 14, and 17.75 GHz. The sweep frequency ranges are 8–12, 11.5–16.5, and 15.5–20 GHz, respectively. Fine-range resolution can be achieved using the 4–5-GHz bandwidth. The SAR image raw data are collected in a "measure-go-stop-measure-go. . ." manner. The step distances of antenna movement were 10, 7, and 7 mm for center frequencies of 10, 14, and 17.75 GHz, respectively. The VNA signal manipulation and the movement of antennas are controlled by software installed on a terminal PC. Three different standard horn antennas are used for the three center frequencies, so after each GB-SAR scan for one center frequency, the antenna should be changed to the corresponding antenna for another center frequency. The disconnection and reconnection of antennas make the repeated recalibration of the SAR system mandatory.

Prior to or after every GB-SAR scan for one frequency, the SAR system was calibrated. The polarimetric calibration of the GB-SAR system is based on the calibration procedure proposed in [22] and Morrison *et al.* [23] with trihedral and dihedrals. The radar cross section (RCS) of the trihedrals was measured in an anechoic chamber, which in this experiment was used as a calibration target.

The calibration was conducted manually and carefully. First, the antennas were aligned vertically downward, checked by a leveling ruler and lifted to a height of approximately 2–4 m. Then, a metal pole was used to mount the trihedrals and dihedrals, and it was placed directly below the antennas, which was realized with the assistance of a plumb bob attached to the center of the four antennas. The metal pole was adjusted manually to ensure that the center of the metal pole was precisely under the pointed tip of the plumb bob when it remained stationary. Then, the plumb bob was removed, and the trihedral was installed on the top of the metal pole. The trihedral was adjusted to point vertically up toward the antennas. We realized this by placing a hard plastic board on top of the trihedral and leveling it. A trihedral has a broad beam pattern such that a sufficiently accurate angular alignment is easily achievable in field conditions (Morrison *et al.* [23]). Then, the RCS of the trihedral is measured by GB-SAR.

For validation purposes, another smaller trihedral is then placed under the antennas, and the RCS measurement is repeated. The calibrated RCS of the smaller trihedral will be compared with the RCS measured in the anechoic chamber to validate the calibration. As described in [22], for cross-polarization calibration, dihedral measurements are needed. The dihedral was rotated by 22.5° from the direction of the rail to produce a high cross-polarization signal. The metal pole was removed, and the RCS of the background was measured. Using the measured trihedral, dihedral, and background RCS mentioned above, the GB-SAR system was calibrated.

After the calibration, the antennas were adjusted to have a 45° pointing angle from vertically down and lifted up to a height of approximately 8.4 m above ground, as shown in Fig. 1. Then, the VV, HH, VH, and HV polarization backscattering were measured sequentially. After SAR images of VV, HH, VH, and HV polarizations at one frequency were obtained, the antennas were changed to another one for the measurement at another band, and the GB-SAR system was calibrated again. For every GB-SAR scan, the height of the antenna was fixed to approximately 8.4 m above the ground, which makes the imaging geometry similar throughout the whole observation season.

Overall, the GB-SAR measurements were conducted daily. Usually, two frequencies among 10.0, 14.0, and 17.75 GHz were measured in one day, and then other frequencies were measured on the next day because of the complicated and careful calibration work, especially in cold environments. Due to holiday, there is a gap in GB-SAR observations from February 11 to February 23.

Using the data processing software, SAR images were formed on the snow surface, with a scene of 10×34 m^2, as shown in Fig. 2(a). The antenna beam pattern is compensated by considering the actual antenna beam pattern and antenna pointing angle. In Fig. 2(a), GB-SAR was installed beyond the top of this image (> -3 m), with the nadir point of antennas at 0 m. At -3 to -27 m in the y-axis, the pure snow surface was observed at different local incidence angles. Far from the GB-SAR at -27 to -37 m in the y-axis, corner reflectors were installed at the boundary between the snow observation field and withered natural high grass; therefore, the image gives a strong backscattering coefficient in this area. The observed sigma0 at different incident angles was extracted from the GB-SAR image in Fig. 2(a) to produce Fig. 2(b). We further applied medium filtering to Fig. 2(b) and extracted sigma0 at certain incident angles for time series analysis.

B. Snow Parameter Measurement

In the experiment, snow and soil parameters were measured in conjunction with the GB-SAR measurement. The soil moisture and temperature were measured automatically using a sensor installed at -3 and -5 cm depths below the soil surface. Snowpits were dug frequently outside the GB-SAR

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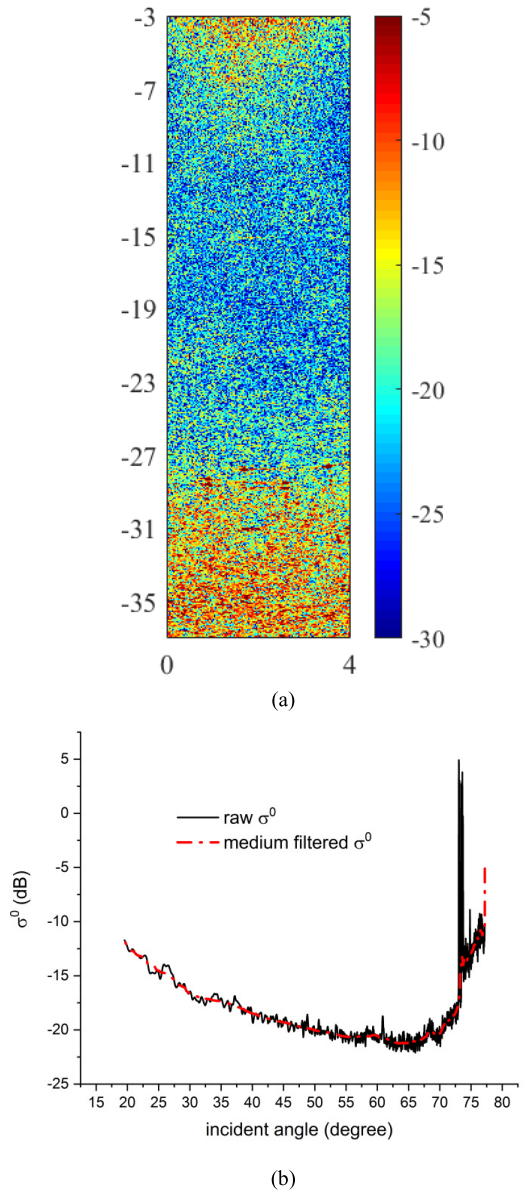


Fig. 2. (a) 17.75-GHz VV-polarization GB-SAR sigma0 image of November 19, 2017, formed on a snow surface with an actual size of 10 × 34 m^2. (b) Extracted mean sigma0 with respect to the incident angle.

field of view, and the snow stratigraphy, snow density, temperature, grain size, and layer thickness were observed following a standard protocol. The snow layering was estimated visually from the pit wall. The snow density was measured by a 250-mL wedge-shaped cutter at 5-cm increments. An additional sample was measured if the remaining thickness on the top was smaller than 5 cm. For example, for snow at a depth of 17 cm, the densities at 0–5, 5–10, 10–15, and 12–17 cm above the soil surface were measured. The snow temperature was measured by a digital thermometer with a tip of 12 cm. The snow grain size was measured by postprocessing the high-resolution photograph of a large number of dispersed crystals over a crystal card of 2-mm reference grids, as shown in Fig. 3. The long axis and short axis of snow grains were digitalized after the photograph was georeferenced using 2-mm grids. An example of digitalization and the statistics of the long axis and short axis snow particle dimensions are shown in Fig. 3. According to the definition, we used the mean value of the long

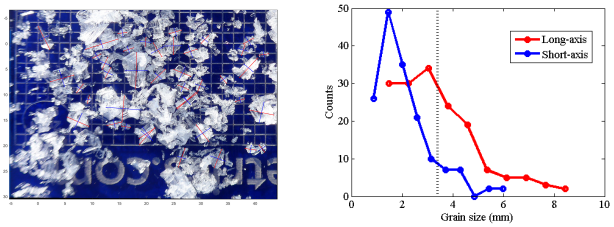


Fig. 3. Snow grain image and its grain size statistics by image processing estimation. The vertical dashed line indicates the mean long axis length.

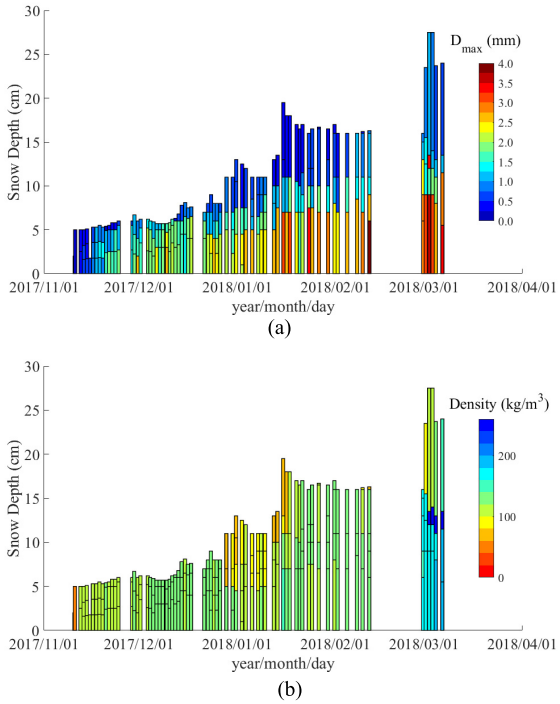


Fig. 4. Snow pit measured (a) time series grain size (Dmax) and (b) density of all the layers.

axis length to represent the geometric grain size (denoted as Dmax) in this study. Fig. 4 shows the 73 measured snowpits used in this study. The time series snow grain size of all the layers is shown in Fig. 4(a), and their corresponding snow density was postprocessed per 5-cm measurement and shown in Fig. 4(b). Significant grain growth can be identified in the time series, especially at the bottom layer.

III. Backscattering Models

In this study, two commonly used snow backscattering models are used to simulate the polarimetric backscattering coefficient of the scene observed by GB-SAR: the first is the dense media radiative transfer theory based on quasicrystalline approximation (DMRT-QCA) model [24], [25] and the second is the DMRT with bicontinuous (DMRT-Bic) model [25]–[29].

The quasi-crystalline approximation (QCA) model is described in [24]. In the DMRT-QCA model used in this study, the snow grains are modeled as sticky ice spheres. The phase matrix, scattering coefficient, and absorption coefficient of the radiative transfer (RT) equation are calculated based on the QCA of Mie scattering of densely packed sticky spheres. In DMRT-QCA, the multiple layer radiative transfer theory is used to simulate the backscattering coefficient.

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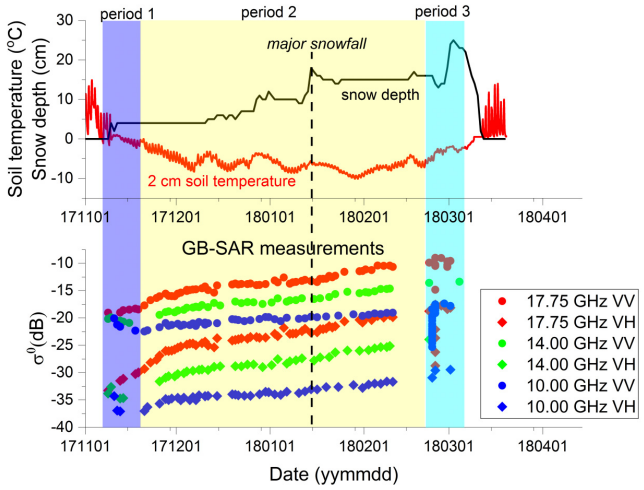


Fig. 5. Time series backscattering of 10.0, 14.0, and 17.75 GHz at VV and VH polarizations measured by GB-SAR, with a 40° incident angle. The snow depth measured by the Altay meteorological site and 2-cm soil temperature are also shown.

In the DMRT-Bic model, the snow microstructure is modeled as a bicontinuous random medium. The modeling of microwave scattering from a bicontinuous medium was described in [28]. Each cell of the bicontinuous medium is approximated as a dipole, and the discrete dipole approximation (DDA) is used to solve the electric field of the bicontinuous medium. Multiple realizations are used to extract the coherent and incoherent fields, and then the phase matrix and scattering coefficient of the bicontinuous medium are calculated using the incoherent field. For multiple-layer snow, when the scattering properties, such as the phase matrix and scattering coefficient of every layer, are calculated, the multiple-layer radiative transfer theory is then used to simulate the backscattering coefficient.

In both the DMRT-Bic and DMRT-QCA models, the multiple-layer radiative transfer equation is solved by the eigenquadrature method. The diffuse intensities are decomposed into Fourier series in the azimuthal direction. Each harmonic is solved by the eigenquadrature approach. After each harmonic is solved, the intensities are composed by Fourier series (see [24], Ding *et al.* [30]).

IV. Results and Analysis

A. GB-SAR Polarimetric Backscattering

From November 2017 to March 2018, the SAR images at the center frequencies of 10.0, 14.0, and 17.75 GHz were obtained repeatedly. The backscattering coefficient was extracted from the calibrated and antenna pattern compensated SAR images. All SAR images were processed, and the time series backscattering at a 40° incident angle is shown in Fig. 5. In Fig. 5, three periods of the time series backscattering coefficient can be identified. In period 1 (November 9–November 20), the backscattering of 10 and 14 GHz shows a decreasing trend, and the backscattering of 17.75 GHz shows a slightly increasing trend. The most likely reason for the time series characteristics of period 1 is a synthetic effect of soil freezing and snow grain growth. The decrease in soil temperature causes a decrease in soil permittivity because the permittivity of frozen soil is very sensitive to the soil temperature in the range of −2 °C to 0 °C due to a significant unfrozen volumetric moisture content change [31]. After November 20, the soil temperature decreased dramatically and remained below −2 °C. In period 1, σ^0 of 10 GHz shows the strongest decreasing trend, which also agrees with the theory that frozen soil permittivity of low frequency is more sensitive to soil temperature than high frequency. For σ^0 at 17.75 GHz because the first snowfall started on November 10, the 17.75-GHz backscattering is most sensitive to snow volume scattering and does not show a decreasing trend. In period 2, all three frequencies show a significant and steady increasing trend caused by the increase in snow depth and snow grain size. For period 3, the backscattering of all three frequencies shows an oscillating pattern caused by the diurnal melt–refreeze cycle of the snowpack.

Our observations indicate that snow grain size can be an important parameter that dominates the snow backscattering change. As shown in Fig. 5, from the start of period 2 to late December, there were no significant snowfall events; however, the backscattering coefficient still increased. Combined with Fig. 4(a), we can conclude that this increase is due to the increase in snow grain size as a result of snow metamorphism. The increasing trend of the backscattering coefficient is stronger in the early stage driven by the smaller snow depth and larger temperature gradient. Later, the increasing speed of the backscattering gradually slowed down, indicating that the snow backscattering signal or the snow grain growth became saturated. The latter assumption agrees with the exponential grain growth models suggested in [32] and [33]. In the middle of January 2018, a heavy snowfall event occurred, as indicated by the dashed line in Fig. 5. After this snowfall, the observed backscattering starts to increase at a higher rate again, especially at 17.75 GHz. This phenomenon indicates that the new snowfall provides more new snow for grain growth. Our hypothesis and explanations mentioned here will be validated later by radar backscattering modeling.

The cross-polarization ratio is the ratio between the backscattering at cross-polarization and co-polarization. In Fig. 5, we can see that the VH backscattering increases more pronounced than the VV polarization in November and December, which indicates that snow volume scattering provides higher cross-polarization signals than soil scattering in this study. This result agrees with a recent study in [34], where the cross-polarization ratio was used for snow depth retrieval in mountains.

In period 3, the microwave backscattering of snow is sensitive to the presence of liquid water. To study the diurnal variation in backscattering caused by the snow melt–refreeze process in detail, an intensive experiment was conducted on February 24–25, 2018, with two diurnal cycles observed and presented in Fig. 6.

In Fig. 6, the air temperature is added as a reference. The backscattering decreased in the daytime due to melting, gradually increased from 14:00 local time in the afternoon, and finally reached saturation at midnight. This phenomenon was also observed in an early experiment [11]. On February 24,

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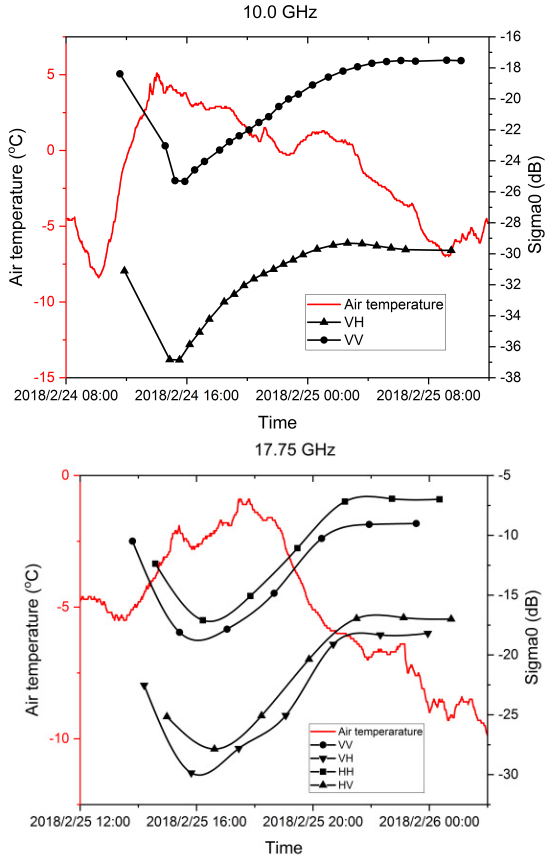


Fig. 6. Diurnal melt–refreeze process of snow and its backscattering coefficient at 10 and 17.75 GHz. The time is 2 h later than local time.

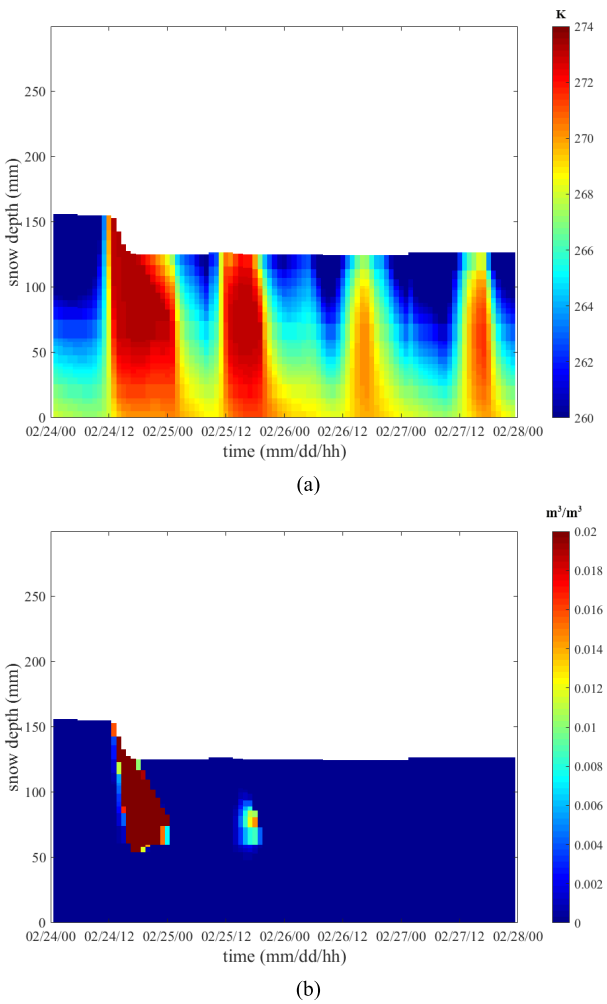


Fig. 7. (a) SNTHERM model simulated snow temperature and (b) volumetric liquid water content during February 24 and 27.

the air temperature varied at approximately 0 °C, but it remained below 0 °C on February 25. However, on both days, diurnal variations in the backscattering at 10.0 and 17.75 GHz were observed, which indicates that liquid water can still exist at negative air temperature. The melting of snow is a result of many factors, such as incoming solar radiation, albedo of snow, longwave radiation from clouds and sky, and latent heat transfer. This process can be simulated by snow process models by considering the mass and energy balance, such as the snow thermal model (SNTHERM) (Jordan [35]). In a previous study, the snow physical process of this experiment was simulated using the SNTHERM model driven by meteorological forcing data [47]. The simulation results show that there was indeed slight snowmelt on February 25 in the snow, as shown in Fig. 7. The snow volumetric liquid water content and snow temperature are shown in Fig. 7, from which we can identify clear snowmelt on February 24 and 25. As shown in [11], a high-frequency backscattering coefficient is more sensitive to snowmelt than a low-frequency backscattering coefficient, which agrees with our observation results that 17.75-GHz backscattering is very sensitive to slight snowmelt events on February 25.

B. Model Simulation

In this article, the DMRT-QCA and DMRT-Bic models were used to simulate the scattering of snow layers, and the advanced integral equation model (AIEM) [36]–[38] and Oh model [39] were used to characterize rough soil surface scattering under snow. The radar backscattering of the snow–soil system was modeled by a multiple-layer radiative transfer scheme in the DMRT-QCA and DMRT-Bic models.

1) Rough Soil Surface Characterization and Modeling: A frozen soil dielectric constant model with soil texture, bulk density, and temperature as inputs was used to simulate the soil permittivity variation caused by the freezing process in November. The soil texture was set as the 11% clay fraction, and the bulk density was 1.9 g/cm^3. The soil temperature was acquired from the sensor-measured temperature at −3 cm depth. The initial volumetric soil moisture was approximately 19% from a ring sampler measurement on November 9. After November 20, the soil temperature was low enough such that the soil permittivity could be set to a constant value that equals the modeled soil permittivity of November 19. The time series soil permittivity used to further support the soil backscattering modeling is shown in Fig. 8.

The AIEM is used to simulate rough soil backscattering. To perform the soil backscattering simulation, the root-mean-square (rms) height and correlation length of the soil surface

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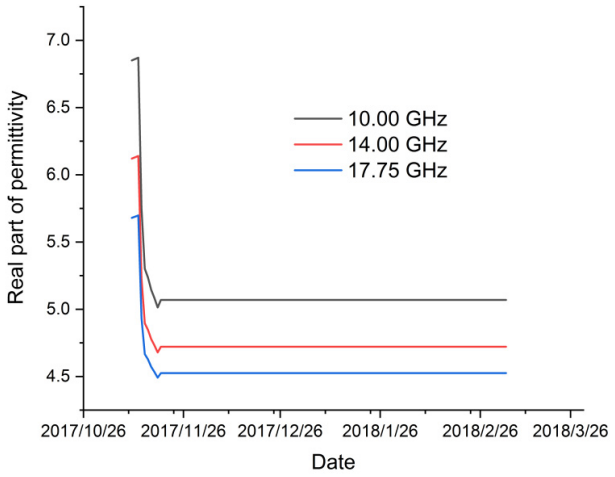


Fig. 8. Simulated time series soil permittivity. Obviously decreasing soil permittivity can be identified in the early stage when the soil temperature is between −2 °C and 0 °C.

in the AIEM were first optimized using the backscattering measurement from the first several days of the experiment, which includes one day with snow-free conditions and three days with thin- and small-grain snow-covered conditions. In the AIEM, an exponential correlation function is used to characterize the rough soil surface. From the optimization using backscattering at copolarization, the rms height and correlation length were fitted as 0.36 and 45 cm, respectively. The same soil parameter was used at all three frequencies. The Oh model [39] is used to simulate the cross-polarization backscattering from bare soil, in which the AIEM co-pol backscattering and a "q" parameter are used. According to Oh *et al.* [39], the "q" parameter is determined by the rms height and Fresnel reflectivity at the nadir of the bare soil surface.

2) DMRT-Bic Model Simulation: In the DMRT-QCA and DMRT-Bic models' simulations, the snow density, temperature, and thickness of each layer can be directly inputted. However, there is a difference between the measured geometric grain size and the snow microstructure metrics required by the models. In DMRT-QCA, snow is modeled as clusters of sticky ice spheres, where an ice sphere radius and a sticky parameter that quantifies the aggregation of ice spheres are needed. In DMRT-Bic, snow is modeled as a bicontinuous random medium, where an optical grain radius and a "b" parameter that describes the grain size distribution are needed [28]. The optical grain radius is defined as the radius of identical spheres that have the same optical properties and equal volume-to-surface ratio as the original nonspherical grains. However, natural snow is not composed of particles of identical size. Chang *et al.* [40] show that the Percus–Yevick models of sticky particle and multiple-size distributions and the bicontinuous models all have tails in the correlation functions that are distinctly different from the exponential correlation function. The tails represent aggregation and stickiness and contribute strongly to microwave scattering. Thus, to completely represent the microstructure of snow in the microwave scattering model, the "b" parameter in the DMRT-Bic model and the stickiness parameter in the DMRT-QCA model are important to represent the tail of the correlation function. However, in this study, only Dmax was measured in the field. Therefore, the method to transfer Dmax to the ice sphere radius (or optical grain radius) needs to be studied, and the "b" parameter in DMRT-Bic and the stickiness parameter in DMRT-QCA need to be determined in an optimization manner. We tested the linear and logarithmic equations to convert Dmax to ice sphere radius, whereas "b" and stickiness parameters were set as constants for all snowpits.

The first attempt is to calculate the optical grain size from Dmax using a simple linear scaling factor. The simple linear relationship between Dmax and optical grain radius tested here is

$$R_0 = D^{\max}/7.0. \tag{1}$$

In the DMRT-Bic model simulation, the soil parameters (including the rms height and soil dielectric constant) are determined by the scheme mentioned above. In both the DMRT-Bic and DMRT-QCA simulations, the snow density measured per 5-cm step was resampled to the mass-weighted average value for each layer, following the stratigraphy visually determined in Fig. 4. The optical grain size of each layer was converted from Dmax using (1). As mentioned above, apart from the optical grain size, another parameter of "b" is needed to quantify the grain size distribution. Theoretically, the grain size distribution of natural snow can vary from case to case. To determine the grain size distribution, the 2-D or 3-D microstructure of snow should be measured by micro-CT, micro-computer-tomography (μCT) scans [41] or section images. However, μCT scans and stereology are both complicated, time-consuming, and require certain instrument support, so they were not conducted in this study. Thus, a constant "b" parameter value was assumed in simulating the time series backscattering.

When the b parameter is set to 1.2, the modeling results are as shown in Fig. 9. We can see in Fig. 9 that on average, the simulated σ^0 agrees well in all nine channels, except that there are some large biases on certain days, which appear randomly in the time series.

To better represent the snow microstructure in the backscattering model, we try to find a more physical relationship between Dmax and optical grain size. The relationship between Dmax and exponential correlation length is complicated, and it has been shown that this relationship is dependent on grain shape [43]. However, we can derive a simple empirical relationship between Dmax and the exponential correlation length (Le) of the snow medium, as shown in [42], which was accomplished using snow stereology images from Mätzler [43]. Following [42], the empirical relationship between Dmax and exponential correlation length can be first determined using snowpit data of Mätzler [43]. A log function is used to describe this relationship between Dmax and the exponential correlation length as

$$\rho_{\text{ex}} = a_1 + a_2 \log(D^{\max}) \tag{2}$$

where the coefficient a_1 is 0.18 and a_2 is 0.09 [38].

If the snow autocorrelation function closely follows an exponential shape, ρ_{ex} equals the correlation length (ρ_c), which

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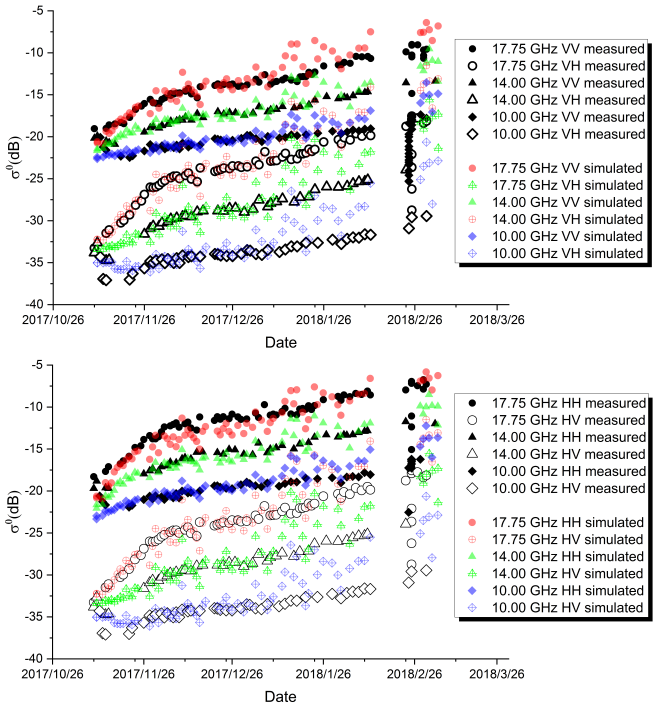


Fig. 9. DMRT-Bic model simulated and GB-SAR measurement of time series backscattering. Linear scaling of Dmax is used.

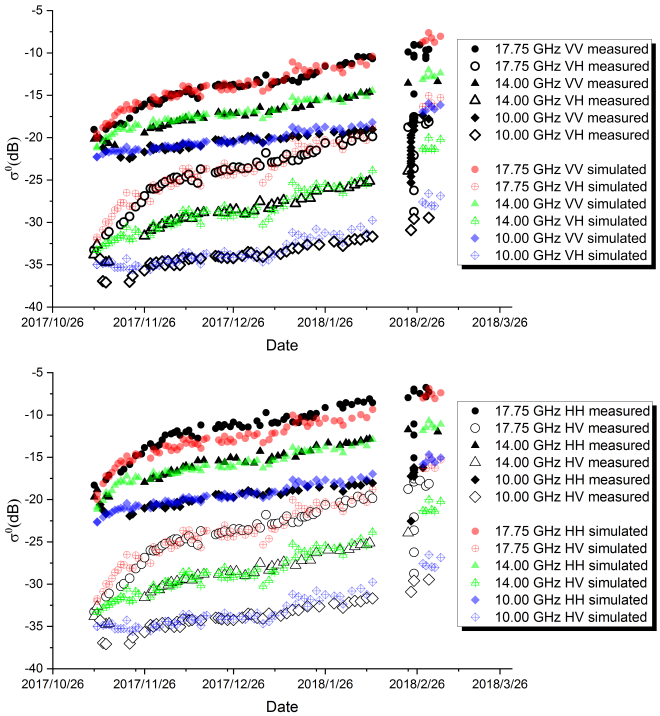


Fig. 10. DMRT-Bic model simulated and GB-SAR measurement of time series backscattering. Logarithm scaling of Dmax is used.

is defined as the inverse of the slope of the autocorrelation function at zero lag distance [43]. In addition, according to the theory of Debye *et al.* [45], the correlation length is theoretically related to the snow specific surface area per ice volume. These two theories give

$$L_c = -\left[\frac{dA(x)}{x}\right]^{-1}\Big|_{x=0} = \frac{4(1-f_v)}{S} \tag{3}$$

where f_v is the volume fraction of ice in snow, calculated as snow density divided by ice density. S is the snow-specific surface area per ice volume in millimeters.

As a result, the relationship between the optical grain size and correlation length can be analytically derived as follows:

$$R_0 = \frac{3\rho_c}{4(1-f_v)}. \tag{4}$$

On the other hand, several studies have demonstrated that a scaling factor β needs to convert ρ_c to ρ_{ex} because few natural snow autocorrelation functions closely follow an exponential shape. Therefore, (4) finally becomes

$$R_0 = \frac{3\rho_{\text{ex}}/\beta}{4(1-f_v)}. \tag{5}$$

We chose a β of 0.75 according to the measurements from Mätzler [43] and Krol and Löwe [46]. The measured Dmax was first converted to the exponential correlation length ρ_{ex} using (2), and then the exponential correlation length ρ_{ex} was converted to R_0 in DMRT-Bic using (5).

It should be mentioned that different scaling relationships between Dmax, optical grain size, and exponential correlation length have been derived in the literature, such as [48], Mätzler and Wiesmann [49], and Brucker *et al.* [50]. In addition, it was found that different exponential correlation lengths can be derived using active and passive microwave observations [48], which indicate that different exponential correlation lengths should be used in active and passive microwave modeling.

Fig. 10 shows the simulated σ^0 based on the logarithmic scaling of Dmax. The soil roughness, soil dielectric constant, and other snow parameters (density, temperature, layer thickness) and "b" were set the same as those used in the linear scaling of Dmax. The results showed that all nine channels of GB-SAR measurements can be well-simulated, including three frequencies and three polarizations. The discrepancy between HH polarization of 17.75 GHz is slightly larger, and the reason is not clear now, which needs further study. By comparing Figs. 9 and 10, we can conclude that much better validation results can be obtained by changing the scaling method for Dmax. The variability of the backscattering coefficient simulation bias is much lower using logarithmic scaling than linear scaling. This is because logarithmic scaling has a stronger theoretical basis, and practically, it suppresses the random observation error for large snow grain sizes.

3) DMRT-QCA Model Simulation: Another commonly used snow backscattering model is the DMRT-QCA model. As described above, apart from the layering and the thickness of each layer, the grain radius, stickiness parameter, and density of snowpack are needed in the DMRT-QCA model to describe snow. The soil parameters can be determined in the same way as described in Section IV-B1).

Similar to the DMRT-Bic model, a conversion from Dmax to ice sphere radius is needed in DMRT-QCA. Here, we also tested two methods: the first is a linear scaling of Dmax directly and the other is a linear scaling of R_0 in DMRT-Bic

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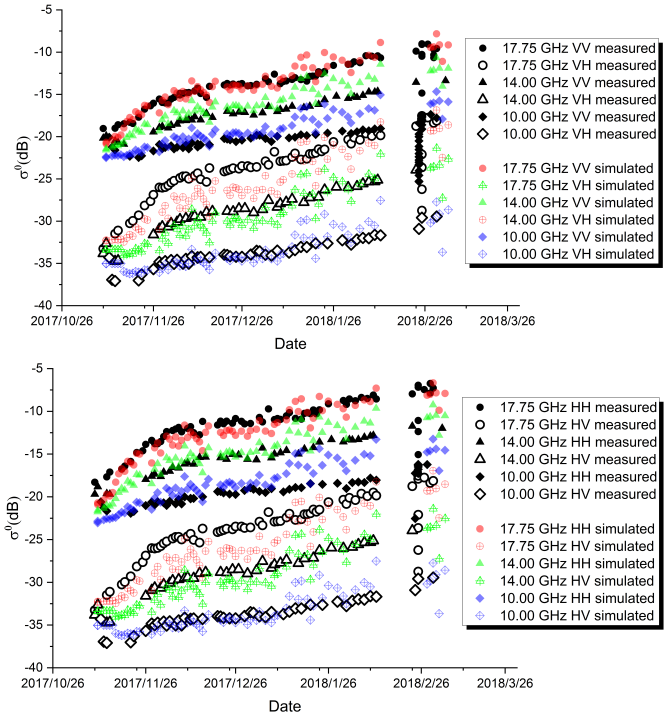


Fig. 11. DMRT-QCA model simulation results with linear scaled Dmax used as grain size.

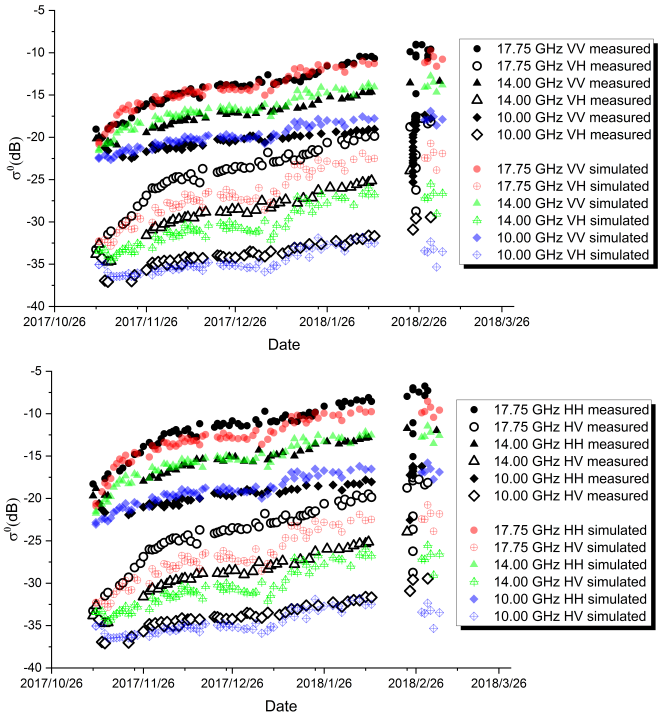


Fig. 12. DMRT-QCA model simulation results with linear scaled optical grain size used as grain size.

based on a logarithmic conversion from Dmax. The stickiness parameter of the DMRT-QCA model is set to 0.1, as suggested in [24].

In this method, the radius of ice spheres in DMRT-QCA is estimated as a simple linear scaling of Dmax as

$$R_{\mathrm{qca}} = 0.16 D^{\max}. \tag{6}$$

The simulation result is shown in Fig. 11. The backscattering coefficient at co-polarization agrees well with the measurement, similar to the results shown in Fig. 8 as predicted by DMRT-Bic. However, the backscattering coefficient using DMRT-QCA at cross-polarization is significantly underestimated at all three frequencies. This is due to the zero cross-polarization elements in the phase matrix of the DMRT-QCA model, which then resulted in lower cross-polarization backscattering predictions [24].

As described in Section IV-B2, the conversion of Dmax can be converted to optical grain size in DMRT-Bic based on a logarithmic scaling method. In the DMRT-QCA model, although the optical grain size cannot be analytically expressed by input parameters, we tested the application of this optical grain size and used a linear scaling factor to convert it to the ice sphere radius (R_{qca}) in DMRT-QCA as follows:

$$R_{\mathrm{qca}} = 1.1 R_0 \tag{7}$$

where R_0 is the optical grain radius in DMRT-Bic calculated by (2) and (5) from Dmax. Parameter 1.1 is the optimized conversion factor. The simulation results are shown in Fig. 12.

In Fig. 12, we can see that the fluctuation in bias of the simulated backscattering in the time series is largely reduced compared with Fig. 11 based on logarithmic scaling from Dmax. Similar to the results in Fig. 10, the DMRT-QCA model still significantly underestimates the cross-polarization backscattering.

A fixed stickiness parameter of 0.1 is used in the DMRT-QCA simulation in this study, but Chang *et al.* [26] proved that stickiness parameters varying from 0.12 to 0.35 for layers of different microstructure can also be used in snow backscattering simulations. This is a two-parameter (grain size scaling coefficient and stickiness parameter) tuning problem. According to our tests, the tuning of both parameters may lead to a slightly better match with observations. According to Chang *et al.* [26], the frequency and grain size dependence of the scattering coefficient is dependent on the stickiness parameter; thus, the stickiness parameter can also be tuned to match the frequency and grain size dependence of observations. Nevertheless, a fixed stickiness parameter of 0.1 can also be used in the DMRT-QCA model to provide a favorable match with observations.

R^2 and RMS error (RMSE) of all the DMRT-Bic and DMRT-QCA model simulations based on different Dmax conversion methods are listed in Table I. Incident angles of 30°, 40°, and 50° are simulated. The DMRT-Bic model with the logarithmic Dmax scaling scheme gives the simulation with the highest R^2 and lowest RMSE at all three incident angles, as indicated by the gray background in Table I. Statistically, we achieved an RMSE of σ^0 below 1.3 dB for all frequencies and polarizations.

4) Snow Volume and Soil Surface Contribution: In the radiative transfer model, the contribution of backscattering from snow volume backscattering and soil surface backscattering can be separated. To better understand the relative

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TABLE I
STATISTICS OF R^2 AND RMSE OF MODEL SIMULATIONS

Incident angle 40 degrees	Model	Linear Scaling		Nonlinear Scaling	
		R^2	RMSE (dB)	R^2	RMSE (dB)
VV	DMRT-QCA	0.93	1.27	0.96	0.97
	DMRT-Bic	0.95	1.17	0.97	0.75
VH	DMRT-QCA	0.94	1.79	0.97	1.94
	DMRT-Bic	0.93	2.17	0.97	1.14
HH	DMRT-QCA	0.92	1.53	0.95	1.16
	DMRT-Bic	0.94	1.34	0.97	1.00
Incident angle 30 degrees	Model	Linear Scaling		Nonlinear Scaling	
		R^2	RMSE (dB)	R^2	RMSE (dB)
VV	DMRT-QCA	0.93	1.09	0.96	0.90
	DMRT-Bic	0.94	1.02	0.97	0.77
VH	DMRT-QCA	0.94	1.54	0.98	1.58
	DMRT-Bic	0.92	1.64	0.97	1.03
HH	DMRT-QCA	0.92	1.33	0.96	1.02
	DMRT-Bic	0.94	1.16	0.97	0.90
Incident angle 50 degrees	Model	Linear Scaling		Nonlinear Scaling	
		R^2	RMSE (dB)	R^2	RMSE (dB)
VV	DMRT-QCA	0.94	1.28	0.96	1.07
	DMRT-Bic	0.95	1.17	0.96	0.92
VH	DMRT-QCA	0.94	2.47	0.97	2.63
	DMRT-Bic	0.93	2.26	0.97	1.26
HH	DMRT-QCA	0.92	1.64	0.95	1.40
	DMRT-Bic	0.94	1.50	0.96	1.31

contribution of snow volume and soil surface at different frequencies, the contributions of snow volume and soil surface backscattering are shown in Fig. 13. The simulation is for a 40° incident angle using nonlinear scaling of Dmax and the DMRT-Bic model. In this study, the radiative transfer equation is solved with the eigenquadrature approach in which multiple scattering is included, and the separation of volume scattering and the volume–surface interaction term is not possible; thus, here, the "volume" term actually means the combination of volume scattering and the volume–surface interaction term.

In Fig. 13, we can see that at 14 and 17.75 GHz, snow volume backscattering is the main contribution. The snow volume backscattering increases dramatically in the early stage (middle and late November).

5) Layering Effect: Natural snowpack tends to form horizontal layers as a result of snow compaction after snowfall events at different times. The backscattering from the multiple layer structure can be simulated using a multiple-layer radiative transfer equation. In the radiative transfer equation resolution, the transmission and reflection on snow layer boundaries are used as the boundary conditions for a multiple-layer snowpack.

One important question arises here: whether the multiple-layer structure plays an important role in radiative transfer and whether a single-layer radiative transfer can predict a similar backscattering based on snow parameters from natural snowpack in this study. Thus, we repeated the DMRT-Bic simulations in Section IV-B2) of the highest accuracy, but a single-layer RT equation is used instead of a multiple-layer RT

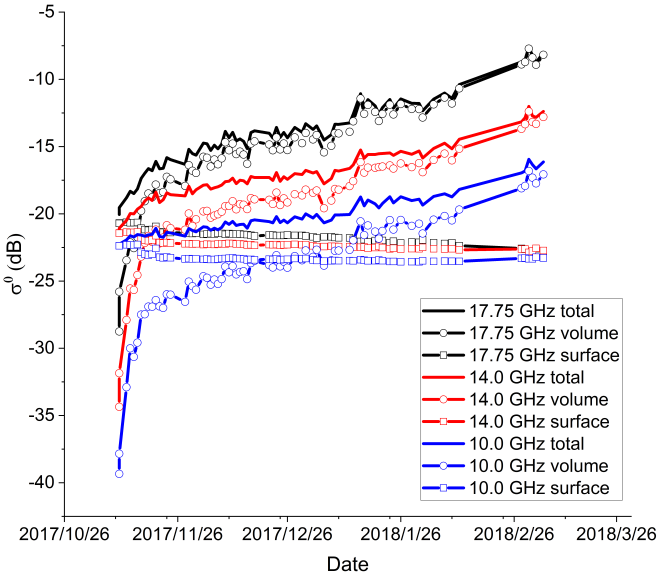


Fig. 13. Contribution of snow volume and soil surface backscattering to total backscattering at 10, 14, and 17.75 GHz simulated by the DMRT-Bic model.

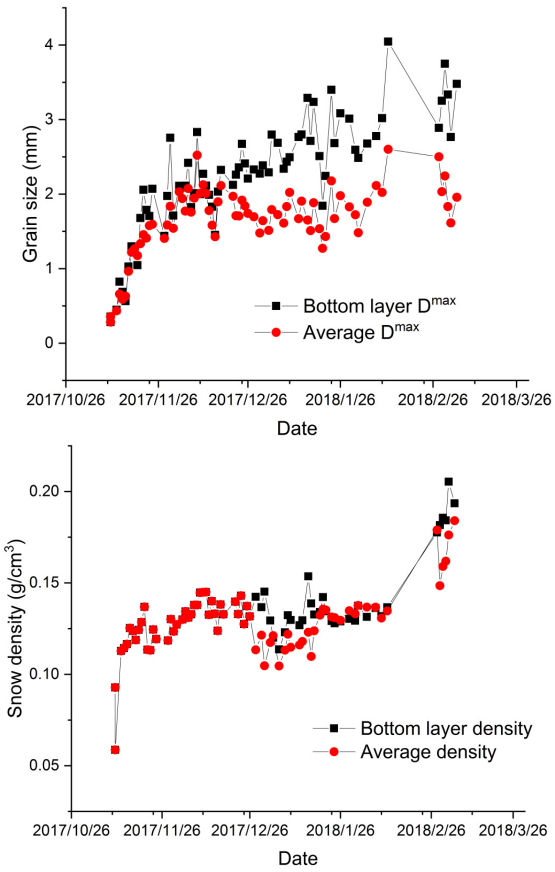


Fig. 14. Time series snowpit-measured average grain size and density in red, with the bottom layer grain size and density in black.

equation. The scaling of Dmax is the same as the nonlinear scaling in Section IV-B2).

In the single-layer RT configuration, the multiple-layer snow density and snow grain size parameter should be aggregated to a single value. In this study, snow density and Dmax are

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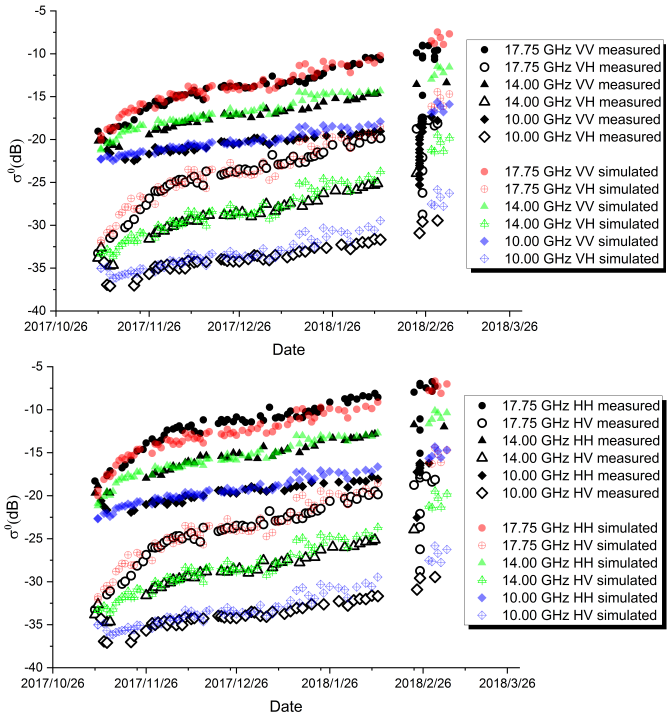


Fig. 15. Single-layer DMRT-Bic model simulation and GB-SAR measurements.

simply calculated by layer thickness weighted averaging. The time series average Dmax and density are shown in Fig. 14, and the backscattering simulation result and its comparison with GB-SAR measurements are shown in Fig. 15. As shown in Fig. 14, the grain size of the bottom layer Dmax grows continuously, similar to an exponential trend as used in [33]; however, the average Dmax is affected by new snowfall and even decreases when significant snowfall occurs.

By comparing Fig. 10 with Fig. 15, we can see that the single-layer RT model can predict a backscattering coefficient similar to that of the multiple-layer RT model, except that there is slight overestimation at VV polarizations of 10.0 and 14.0 GHz after middle January, immediately after a major snowfall event. The performance of the single-layer simulation indicates that a single layer may be used in the snow parameter retrieval algorithm by ignoring the multiple-layer structure. It should be noted that the conclusion here is made based on shallow snow observations and simulations; in our experiment, the maximum snow depth was no larger than 30 cm. In addition, the HH polarization can be more sensitive to layering due to scattering on the layer boundaries. However, the permittivity contrast of each layer in this study is very small due to the small density in general (50–220 kg/m^3) observed in the experiment; thus, the difference between the single-layer and multiple-layer RT model simulations is small. For multiple-layer snow with significant permittivity contrast between layers, the difference in single- and multiple-layer model predictions would be expected to be larger.

V. Conclusion

In this article, a one-winter experiment to observe the time series backscattering coefficient of snow-covered bare soil is reported. This is the first long time series snow-covered soil backscattering experiment that was measured by an imaging radar. The backscattering coefficient was observed at three frequencies covering the X-band and dual-Ku bands, which are of great interest to the snow remote sensing community used for SWE estimation in mountains. The GB-SAR system was calibrated manually, carefully, and frequently. Once the antenna was changed to conduct the measurement at another frequency, calibration was performed immediately before or after the successful normal scans at four polarizations to reduce the temperature drift. The snow and soil parameters were measured in conjunction with the backscattering measurement and were used for the DMRT-QCA and DMRT-Bic snow backscattering models to test their efficacy in reproducing the GB-SAR observations.

The observations from our experiment show that in general, the time series backscattering signature of snow-covered terrain can be classified into three periods, and its variation is mainly driven by the soil freezing, snow grain size growth, and snow accumulation processes. In the first period, the surface soil temperature drops from approximately 0 °C to approximately −2 °C. A decreasing trend of the backscattering coefficient can be observed at 10.0 and 14.0 GHz, whereas an increasing trend can be observed at 17.75 GHz. In the second period, the permittivity of the surface soil was stable, and the backscattering was mainly affected by snow properties. A significant increase in backscattering can be observed at all channels, especially at high frequencies. In the third period, melt–refreeze of snow occurred, and the backscattering showed diurnal variation.

The time series observations mentioned above for dry snow (basically period 1 and period 2) can be modeled by backscattering models. The DMRT-Bic model provides very good agreement with measurements for frequencies and polarizations, whereas DMRT-QCA underestimates the VH polarization due to the zero element in the phase matrix it used. In model simulations, linear and nonlinear conversion of snow-pit measured geometric grain size (Dmax) to optical grain radius in DMRT-Bic and ice sphere radius in DMRT-QCA model were tested; there is a difference between the observed and model-used snow microstructure parameters. The results showed that the nonlinear conversion of Dmax gives a better simulation performance and practically reduces the observation error in the larger grain size range. Based on our simulations, we suggest using the DMRT-Bic model based on a nonlinear conversion of Dmax for both polarimetric snow backscattering modeling and snow parameter retrieval purposes. A lookup table or a semiempirical parameterization may be needed to reduce the computational cost.

In this study, we also tested the single-layer configuration compared with the original multiple-layer configuration in the DMRT-Bic simulation. The snow grain size and density were averaged using layer thickness. Our experiment with a maximum snow depth smaller than 30 cm shows that the single-layer model can also reproduce the observed snow backscattering, indicating that a single-layer DMRT-Bic model may be used in retrieval algorithms. Further study is needed to check its validity for a thick snowpack.

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